

BRYAN KUMARA

<https://brykumara.github.io/>

PERSONAL INFORMATION

Place of Birth: Singapore

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BIOGRAPHY

I am currently a PhD student at the University of Oxford, advised by Prof. Kasper Rasmussen focusing on post-quantum cryptography in privacy-preserving search. Before starting my PhD, I was an Applied Cryptographer at CrossBar Inc. working on threshold ECDSA and a Research Assistant at the Alan Turing Institute working on privacy enhancing technologies. In the latter, I co-authored the Mahi-Mahi asynchronous DAG consensus protocol, BACON - a batched vector commitment for MPCitH-style signatures, and secure guidelines for implementing homomorphic encryption schemes.

EXPERIENCES

- 2025 - Applied Cryptographer, CrossBar Inc., United States
Threshold ECDSA and Post-Quantum Transition
- 2023 - 2025 Research Assistant, The Alan Turing Institute, United Kingdom
Asynchronous Consensus, Batched Vector Commitments, Homomorphic Encryption
- 2021 - 2021 Project Management Intern, PayPal, United Kingdom
Brexit Project
- 2019 - 2019 Undergraduate Researcher, Imperial College London, United Kingdom
Model Checking for Multi-Agent Systems
- 2018 - 2018 Project Intern, Boston Consulting Group, Indonesia
Timber and Tissue Paper Manufacturing in Indonesia
- 2017 - 2017 Research Assistant, The Habibie Center, Indonesia
Cybersecurity Policies in Asean

EDUCATION

- 2025 - University of Oxford, Oxford, United Kingdom
DPhil in Applied Cryptography
- 2021 -2022 University College London, London, United Kingdom
M.S in Emerging Digital Technologies
- 2017 - 2020 London School of Economics, London, United Kingdom
B.S in Philosophy, Logic, and Scientific Method

PUBLICATION, ACADEMIC WORK, AND AWARDS

My full list of publications, academic work, and awards is available at <https://brykumara.github.io/>